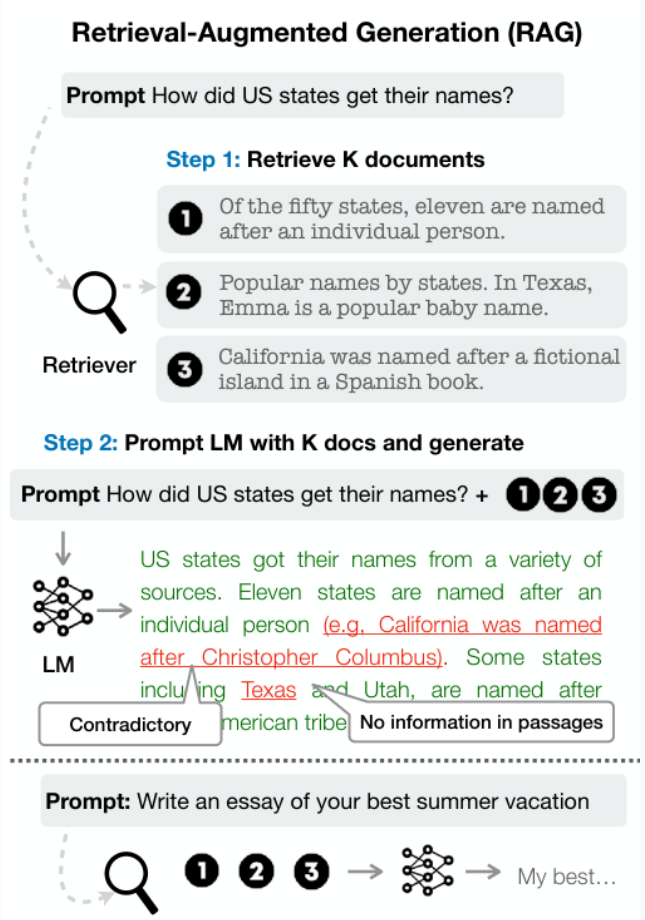
Self-RAG: A Language Model That Retrieves, Generates and Criticises Its Own Answers

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Based on: A. Asai, Z. Wu, Y. Wang, A. Sil, H. Hajishirzi. "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection". ICLR 2024

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The Problem with Standard RAG



Conventional RAG: always retrieves K documents and generates without checking them. Source: Asai et al., ICLR 2024, Figure 1 (left).

- LLMs write fluent answers but make factual mistakes
- **RAG:** find documents and add them to the model input
- Standard RAG always retrieves K documents, even when they are not needed
- It never checks whether the answer agrees with the documents

Outline

1. The main idea of Self-RAG
2. Training the model
3. Experiments and results
4. What this means for reliable language models

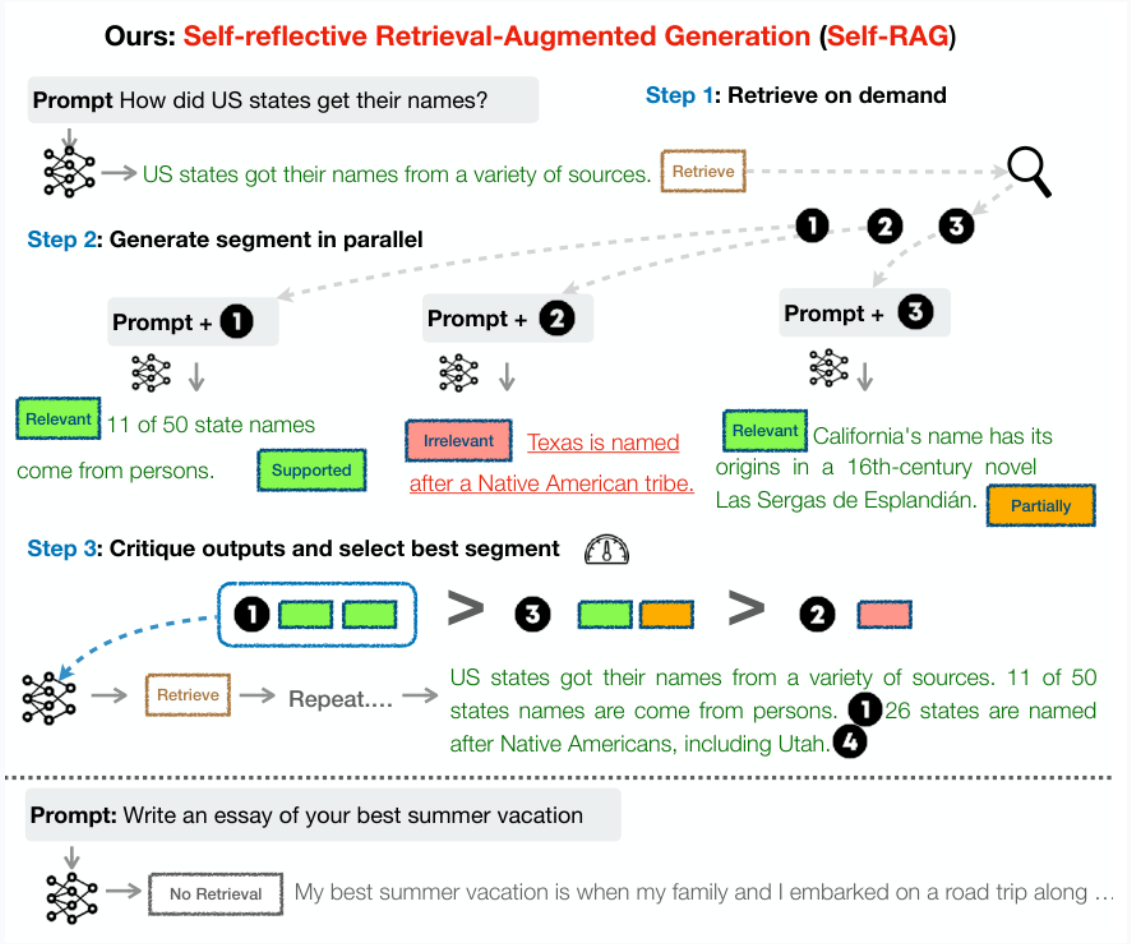
Reflection Tokens

One model: retrieve when needed, generate, criticise its own output

Token Name	Values	Question
Retrieve	{yes, no, continue}	Should the model search?
IsRel	{relevant, irrelevant}	Is the passage relevant?
IsSup	{fully supported, partially supported, no support}	Is the output supported by the passage?
IsUse	{5, 4, 3, 2, 1}	How useful is the whole response?

Self-RAG at Work

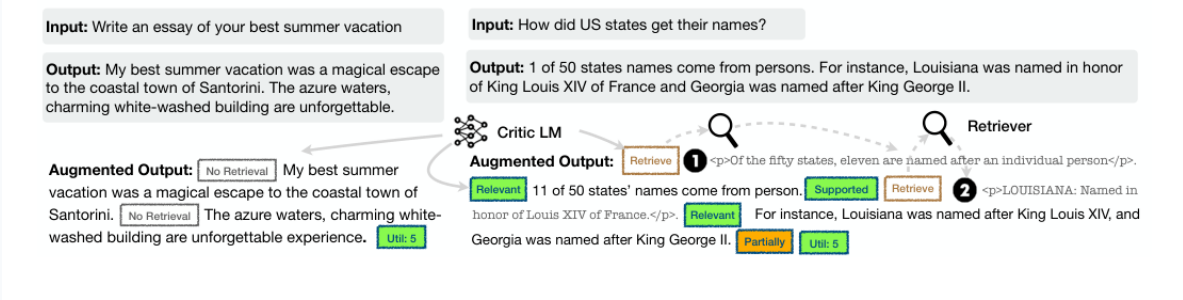
- **"How did US states get their names?":** Retrieve, then 3 passages
- A continuation for each passage, generated in parallel
- Critique tokens select the relevant and fully supported segment
- **"Write an essay of your best summer vacation":** no retrieval



Self-RAG retrieves on demand, generates segments in parallel and selects the best one. Source: Asai et al., ICLR 2024, Figure 1 (right).

Training: Critic and Generator

1. **GPT-4 labels:** 4k–20k examples for each token type
2. **Critic model trained on these labels:** over 90% agreement with GPT-4 for most token types
3. The critic inserts reflection tokens into 150k training examples
4. **Generator (Llama 2, 7B or 13B)** learns text and tokens together; no critic at test time



Self-RAG training examples: without retrieval (left) and with retrieved passages (right). Source: Asai et al., ICLR 2024, Figure 2.

Control at Test Time

The model writes the reflection tokens itself — no retraining needed



Weights

e.g. a higher weight on IsSup means more focus on support



Threshold

Retrieve more or less often

Six Evaluation Tasks

Short-form QA

- PopQA
- TriviaQA

Closed-set

- PubHealth (fact checking)
- ARC-Challenge (reasoning)

Long-form

- Biography generation
- ASQA (answers with citations)

Main Results: Self-RAG 13B vs ChatGPT

Task and Metric	Self-RAG 13B	ChatGPT
PopQA (rare entities), accuracy	**55.8**	29.3
PubHealth (health fact checking), accuracy	**74.5**	70.1
ASQA citation precision	**70.3**	65.1 (with retrieval)

Source: Asai et al., ICLR 2024, Table 2

Why Reflection Tokens Matter

PopQA accuracy — Llama2-FT 7B: same training data, no reflection tokens

48.7

Llama2-FT 7B

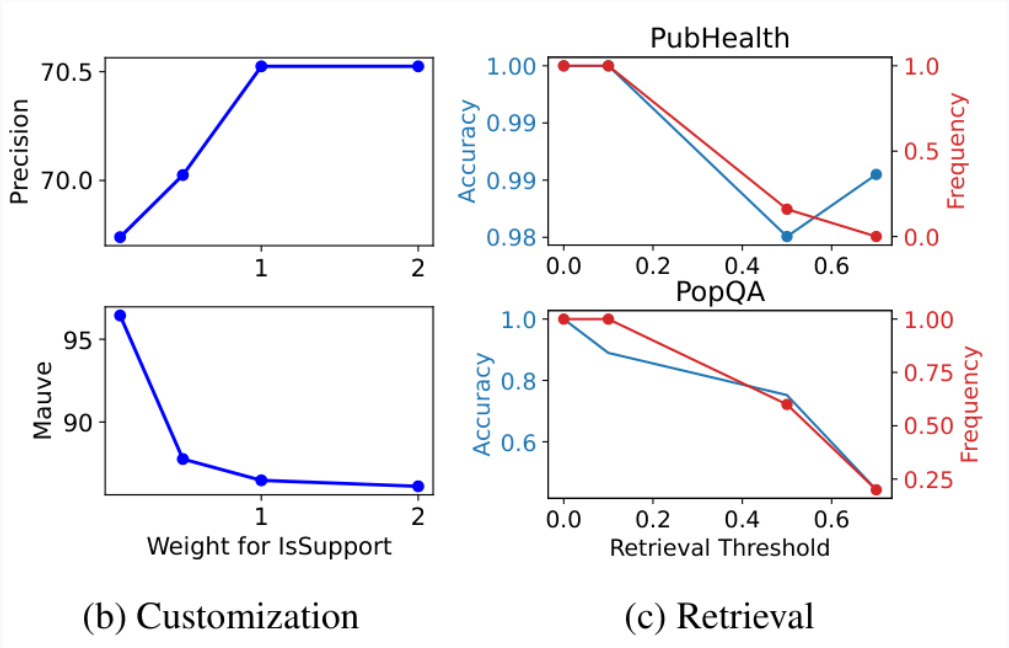
54.9

Self-RAG 7B

The gain comes from the method, not only from the data

Source: Asai et al., ICLR 2024, Table 2

Trade-offs



Effect of the IsSup weight on ASQA (b) and of the retrieval threshold on PubHealth and PopQA (c). Source: Asai et al., ICLR 2024, Figure 3 (b, c).

- **Higher IsSup weight:** higher citation precision, lower fluency (MAUVE) on ASQA
- **Higher retrieval threshold:** fewer searches; accuracy drops on PopQA more than on PubHealth

Conclusions

1. One model decides when to search and checks its own answers with reflection tokens
2. Open 7B and 13B models outperform ChatGPT on several tasks and give more precise citations
3. Behaviour can be tuned at test time; claims are still not always fully supported

Key Message

"A language model becomes more reliable when it learns to check its own answers against its sources"

Thank you! Questions?